

Abstract problem solving and discrete math preview

What do I mean by abstract problem solving

So far most of the problems we've solved were very visual in nature, or were gamified in nature

Most problems you'll solve in programming is more *logical* and so the ability to

solve problems with pure logic and values

become important

What is discrete math

Discrete math is a subject you may (or may not) have later on, but in essence it's a branch of math that

Studies distinct *structures* with a focus on logic, proofs, sets, and combinatorics

I'll be introducing some of these concepts early, to help us get a better grasp of logical thinking

Propositions and connectives

- Paris is the capital of the Philippines
- Ten is greater than seven
- Every even number is the sum of two prime numbers

A **proposition** is simply a *statement* that is either *true* or *false*

note that you don't have to *know* if it's true or false, only that it's answerable by true or false

What is the **truth-value** of the first proposition

Propositions

- Where are you going?
- Close the door
- This statement is false

These are examples of statements that **are not propositions**

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The third statement is considered a paradox, which means we cannot give it a *truth-value* so we ignore it in discrete math

Propositions

- Jon is tall
- Chocolate is delicious
- $x > 10$

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These are also **not propositions**

Connectives

There are things you can connect to a proposition called *connectives*

1. negation $\neg$
2. and $\wedge$
3. or $\vee$
4. if-then (implies) $\implies$
5. if-and-only-if (equivalent) $\iff$

There are other forms of these symbols, and other names, but they are logically the same

Exercise

Using the proposition

p = "Today is Monday" and q = "It is raining"

Write the following logical expressions in words

1. $\neg q$ = It is **not** raining
2. $p \wedge q$
3. $\neg p \wedge q$
4. $p \vee q$
5. $p \vee \neg q$
6. $p \implies q$ = **if** today is monday **then** it is raining
7. $\neg p \implies q$
8. $p \iff q$
9. $\neg p \iff q$

Tables and compound values

